

Master Technical Skills & Applied Engineering Systems Compendium

An exhaustive technical compendium indexing **38 core engineering skills, mathematical algorithms, and architectural patterns** engineered across 28+ custom production codebases and distributed tools. Demonstrates verified production mastery in **Autonomous AI Architectures, LLVM Compilers & Custom Language Toolchains, C++20 AR/VR Spatial Game Engines, Zero-Knowledge Cryptography, Computer Vision Biometrics, and Enterprise Full-Stack Platforms.**

- ⚡ Low-Level Systems & LLVM
- 🧠 Autonomous Cognitive AI
- 👤 C++20 Spatial Engine & ECS
- 🔒 Zero-Knowledge Cryptography
- 👁️ 3D Computer Vision Biometrics
- 🌐 High-Concurrency Cloud APIs

CORE ENGINEERING DISCIPLINES & COMPETENCIES MATRIX

| ENGINEERING DISCIPLINE                     | APPLIED COMPETENCIES & TECHNICAL STACK DEMONSTRATED IN TOOLS                                                                                                         |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Autonomous AI & Cognitive Systems       | Cognitive Routing, Intent Decomposition, Autonomous Executive Controllers, Synthetic DNA Evolution, Multi-Tiered Memory, Safety Kernels, Quantized Local Inference.  |
| 2. Compilers, Language Design & Systems    | Lexical Tokenization, Recursive Descent AST Parsing, LLVM IR Codegen, Static Type Systems, Custom Standard Library (‘omlib’), SIMD Cache-Aligned Structs.            |
| 3. Spatial Computing, AR/VR & Game Engines | C++20 Entity Component System (ECS), High-Concurrency UDP Networking, Client Prediction, HLSL Shaders, Deterministic 3D Physics, Cross-Engine Bridging.              |
| 4. Computer Vision, Biometrics & ML        | MediaPipe 3D Face Mesh, Eye Gaze Vector Estimation (EAR), Head Pose Telemetry (SolvePnP), NLP TF-IDF Threat Classifiers, ROC-AUC Anomaly Tuning.                     |
| 5. Cybersecurity, Cryptography & Defensive | AES-256-GCM Zero-Knowledge Vaults, PBKDF2 Key Derivation, Shannon Entropy Auditing, Port/CVE Auditing, Hardened JWT/Bcrypt Gateways, Memory-Safety Invariants.       |
| 6. OSINT, Graph Analytics & Intelligence   | Open-Source Intelligence Footprinting, Recursive Semantic Crawlers, Multi-Dimensional Knowledge Graphs, Social Network Analysis (SNA), Louvain Clustered Topologies. |
| 7. Full-Stack Web, APIs & Creative UI/UX   | Next.js 14+ (SSR/SSG), React 18, Strict TypeScript, FastAPI Asynchronous Services, PostgreSQL CRM Architecture, Three.js WebGL & GSAP Shaders.                       |
| 8. Automation, Network Infra & DSP         | Manifest v3 Chrome Extensions, Bezier Human Emulation, Rotating Proxy & JA3/JA4 TLS Spoofing, Background Daemons, MIDI DSP & python-pptx AI Pipelines.               |



### Cognitive Intent Routing & Task Decomposition

Python Intent Classifiers Task Trees Adaptive Compute Latency Budgets

Engineered autonomous routing engines that dissect natural language queries into hierarchical task graphs. Dynamically provisions compute budgets and selects optimal reasoning modes (Fast vs. Multi-Step Deep Reasoning) based on algorithmic difficulty scores, eliminating token waste and maximizing latency efficiency for complex problem-solving workflows.

### Autonomous Executive Controllers & Session Orchestration

State Machines Execution Sandboxes Tool Calling Step Verification Audit Trails

Architected deterministic executive controllers that govern agent lifecycles, execution sandboxes, and multi-tool orchestration. Implemented runtime step verification, hard timeout budgets, and rollback mechanisms, ensuring autonomous agents execute multi-step tool calls with zero uncontrolled side-effects and generate cryptographically verified audit trails of every reasoning step.

### Synthetic Heuristic Evolution & Experience Distillation

Genetic Algorithms Experience Distillation Checkpointing SHA-256 Gene Hashes

Designed self-learning cognitive engines that analyze runtime execution traces, isolate successful reasoning heuristics, and distill them into persistent synthetic heuristic genes. Implemented hash-verified DNA evolution pipelines and state checkpointing, enabling autonomous systems to adapt strategies across sessions and continuously reduce repetitive problem-solving overhead.

### Multi-Tiered Memory Systems (Episodic & Semantic)

Vector Embeddings Semantic Retrieval SQLite Episodic Replay Context Gating

Developed hybrid memory architectures unifying episodic execution logs, semantic vector retrieval, and procedural stores. Engineered memory gating and context-injection pipelines that retrieve historical interaction patterns and inject relevant contextual priors into live reasoning loops without overwhelming context windows or causing associative hallucination.

### Continuous Daemon Scheduling & Energy Governance

Async Daemons Background Scheduling Resource Governors OS Process Hooks

Implemented continuous background runtimes and hardware-aware energy governors to orchestrate long-horizon autonomous tasks. Engineered resource-aware scheduling daemons that throttle computational throughput, monitor system utilization, dynamically adapt priorities during idle windows, and safely trigger maintenance cycles, state persistence, and memory compaction.

### Bounded Safety Kernels & Operational Policy Enforcement

Sandboxing RBAC Read-Only Guards Network Isolation Multi-Level Trust

Engineered deterministic safety kernel layers that validate incoming commands, enforce active operational constraints (read-only access, network isolation, destructive-write guards), and restrict tool execution boundaries. Built multi-level trust verification mechanisms to ensure autonomous code execution adheres strictly to user-defined safety policies before applying system modifications.

### Model Runtime Optimization & Local Inference Pipelines

Model Registry Quantized GGUF/ONNX Fallback Cascades Response Streaming

Built modular model registry and runtime abstraction layers capable of dispatching tasks across cloud APIs and quantized local inference endpoints. Implemented automatic fallback cascades, token rate-limiting, and response streaming protocols, ensuring uninterrupted model availability and high-throughput inference under volatile network conditions.

### Lexical Tokenization, Grammar & AST Parser Engineering

Custom Lexer Recursive Descent Parsing Abstract Syntax Trees  
Syntax Diagnostics

Designed custom programming language frontends featuring deterministic lexical analyzers and recursive descent AST parsers. Developed formal grammar specifications capable of processing custom syntax structures, operators, control flow statements, and nested block scopes while providing line-accurate compiler diagnostics, syntax error recovery, and structured AST representations.

### LLVM Intermediate Representation (IR) & Native Codegen

LLVM C++ API Target Machine Passes Native Code Emitters  
JIT Compilation

Constructed compiler backends utilizing LLVM infrastructure to translate custom high-level Abstract Syntax Trees into optimized LLVM Intermediate Representation (IR). Implemented instruction emission, register allocation passes, target machine configurations, and native object code generation pipelines, producing high-performance standalone machine binaries from scratch.

### Static Type Systems & Semantic Verification

Symbol Tables Type Inference Scope Resolution  
Semantic Analysis Passes

Implemented static type-checking and semantic verification engines for custom compiled languages. Designed lexical symbol tables, variable scope resolvers, and type coercion validators that enforce strict compile-time type safety, eliminate undefined symbol access, and optimize memory layout prior to native binary emission.

### Custom Standard Library & Native Memory Primitives

Manual Memory Mgmt I/O Buffers Core Math Primitives  
C ABI Interoperability

Authored custom standard libraries and core runtime primitives to support native language execution. Engineered low-level memory allocation wrappers, string manipulation buffers, terminal I/O routines, and mathematical primitives. Implemented standard C ABI linking interfaces, enabling custom-compiled programs to invoke operating system system-calls and native external shared libraries.

### SIMD Vectorization & Cache-Aligned Data Structures

Data-Oriented Design Cache Line Alignment Memory Padding  
C++20 SoA Layout

Applied low-level hardware optimization techniques to maximize CPU cache locality and instruction throughput. Structured memory buffers along 64-byte cache lines, utilized struct-of-arrays (SoA) layouts to facilitate SIMD auto-vectorization, and eliminated pointer chasing in critical execution paths, achieving significant throughput speedups in compute-intensive loops.



## Spatial Computing, AR/VR & Game Engine Architecture

5 Competencies

### Entity Component System (ECS) Architecture

C++20 Data-Oriented Design Cache Locality  
Archetype Pooling Parallel Systems

Architected high-performance, multithreaded Entity Component System (ECS) game server architectures in modern C++20. Applied data-oriented design principles, contiguous memory archetype storage, and component pooling to eliminate object-oriented cache misses, enabling real-time parallel system updates for thousands of simultaneous entities across high-frequency simulation ticks.

### Low-Latency Network Synchronization & State Replication

UDP Sockets Delta Compression Dead-Reckoning  
Client-Side Prediction

Engineered authoritative client-server spatial networking systems built on raw UDP socket protocols. Implemented delta-compression algorithms, client-side movement prediction, and dead-reckoning interpolation to guarantee synchronized spatial positions across distributed clients, minimizing bandwidth utilization and eliminating visual stutter across variable network connections in multiplayer virtual environments.

### HLSL Shader Programming & Unity Rendering Pipelines

Unity C# HLSL Universal Render Pipeline (URP)  
Custom Shaders Spatial FX

Developed custom HLSL vertex and fragment shaders integrated within Unity's Universal Render Pipeline (URP). Created interactive spatial visual effects, holographic depth cues, spatial boundary grids, and real-time lighting materials optimized specifically for high-frame-rate rendering across virtual and augmented reality head-mounted displays.

### Cross-Platform Spatial Engine Bridging

Protocol Translators TCP/WebSockets  
Minecraft & Roblox Bridges Serialization

Engineered bidirectional cross-platform bridge protocols linking standalone C++ spatial simulation engines with external game platforms such as Minecraft and Roblox. Implemented serialization protocols, entity mapping wrappers, and real-time WebSocket state broadcasters, allowing spatial avatars and world modifications to mirror continuously across disparate gaming engines.

### Deterministic 3D Spatial Physics & Collision Simulation

AABB Bounding Boxes Raycasting Spatial Partitioning  
Fixed-Step Physics

Implemented deterministic 3D physics engines featuring axis-aligned bounding box (AABB) collisions, raycasting, and spatial hashing grids. Maintained fixed-timestep simulation loops to ensure identical physics resolution across heterogeneous client hardware, preventing desynchronization in competitive spatial multi-user simulations.



## Computer Vision, Biometrics & Machine Learning

4 Competencies

### Facial Mesh & Eye Gaze Vector Estimation

OpenCV MediaPipe Face Mesh Iris Tracking 3D Geometry  
Euclidean Vectors

Implemented real-time facial landmark tracking and gaze estimation pipelines using OpenCV and MediaPipe. Extracted 468 3D facial coordinate landmarks to calculate Euclidean Eye Aspect Ratios (EAR) and iris center displacements, accurately estimating user eye gaze direction and blink frequency at 60 FPS without specialized hardware sensors.

### Biometric Attention & Posture Telemetry

SolvePnP 3D Head Pose Real-Time Telemetry  
Ergonomic Analytics Fatigue Detection

Built automated biometric tracking engines calculating 3D head pose rotation vectors (pitch, yaw, roll) using OpenCV's SolvePnP algorithm. Formulated attention scoring heuristics cross-referencing gaze stability, head orientation, and eye closure durations to detect distraction and fatigue, generating automated alerts and rich ergonomic productivity analytics.

### NLP Feature Extraction & Threat Classification

Scikit-Learn TF-IDF Vectorization N-Grams Regex Heuristics  
Ensembles

Developed natural language processing pipelines to detect deceptive phishing emails and malicious messaging. Extracted statistical lexical features, suspicious URL structural patterns, domain spoofing heuristics, and TF-IDF word/character n-grams, applying ensemble machine learning classifiers to accurately categorize sophisticated social engineering attacks and automated spam campaigns.

### Statistical Anomaly Detection & Model Evaluation

Confusion Matrices ROC-AUC Curves Precision-Recall Tuning  
Scikit-Learn

Engineered rigorous machine learning validation workflows evaluating model performance across imbalanced security datasets. Tuned classification thresholds, optimized ROC-AUC curves, and applied cross-validation to minimize false-positive rates in production security screening, ensuring legitimate high-priority communications remain unhindered.

### AES-256-GCM Cryptography & Zero-Knowledge Vaults

AES-256-GCM PBKDF2 HMAC-SHA256 Secure Memory Scrubbing  
SQLite Security

Architected zero-knowledge encrypted credential vaults utilizing authenticated AES-256-GCM encryption and PBKDF2 HMAC-SHA256 key derivation with high iteration counts and cryptographic salts. Implemented secure in-memory key handling with automatic memory scrubbing, tamper-resistant SQLite database encryption, and automated clipboard sanitation to prevent memory dump exploitation.

### Password Strength & Cryptographic Entropy Calculation

Shannon Entropy Zxcvbn Spatial Pattern Analysis  
Brute-Force Estimation

Engineered real-time password security analyzers calculating information-theoretic Shannon entropy and heuristic pattern vulnerability. Evaluated dictionary permutations, spatial keyboard walks, sequential repetitions, and estimated offline brute-force cracking durations, delivering dynamic visual feedback and tailored suggestions to enforce resilient, entropy-dense passphrases resistant to modern GPU cracking.

### Automated Port Auditing & CVE Vulnerability Mapping

Socket Programming Multithreading Banner Grabbing  
CVE Databases Auditing

Developed multi-threaded network reconnaissance tools in Python for automated infrastructure auditing. Performed concurrent TCP/UDP port scans, service identification, and raw banner grabbing, automating the cross-referencing of discovered daemon versions against published CVE vulnerability databases to deliver prioritized risk scores and actionable technical remediation summaries.

### Hardened Authentication & API Gateway Security

JWT / OAuth2 Bcrypt Salted Hashing Rate Limiting RBAC  
CSRF/XSS Defense

Engineered enterprise-grade authentication gateways featuring Bcrypt password hashing, short-lived JWT access tokens with rotating refresh tokens, and Redis-backed IP rate-limiting. Enforced strict HTTP-only secure cookie policies, CSRF token validation, and granular role-based access control (RBAC), effectively eliminating brute-force attacks and session hijacking vulnerabilities.

### Custom Cryptographic Stream Obfuscation & Bitwise Ciphers

Bitwise Permutation Dynamic S-Boxes Stream Ciphers  
Binary Masking

Designed experimental cryptographic obfuscation algorithms and lightweight stream cipher prototypes. Implemented multi-round bitwise rotations, dynamic S-box substitution matrices, and key-dependent byte shuffling to obscure sensitive binary payloads, evaluating cipher resistance against statistical frequency analysis, pattern detection, and automated decompilation during static reverse-engineering analysis.

### Defensive Memory Safety & Anti-Tamper Invariants

Buffer Boundary Checks Secure Zeroing Cryptographic Hashes  
Integrity Checks

Implemented defensive systems programming techniques that prevent buffer overflows, use-after-free conditions, and memory corruption. Built tamper-evident integrity checkers that continuously verify executable binary hashes, preventing code injection, DLL hijacking, and unauthorized runtime memory patching.

## OSINT, Graph Analytics & Web Intelligence

### Open-Source Intelligence (OSINT) & Digital Footprinting

Automated Scraping DNS Enumeration Metadata Mining  
Dossier Aggregation

Developed modular OSINT intelligence tools automating entity footprinting across public digital channels. Scripted concurrent scrapers and API aggregators for multi-platform username validation, email deliverability probing, DNS domain tracking, and file metadata extraction, synthesizing disparate unstructured OSINT data points into unified investigation dossiers for security researchers.

### Recursive Semantic Crawling & Knowledge Graphs

Graph Theory Recursive Crawlers NetworkX D3.js  
Vector Nodes Topologies

Engineered automated research engines that recursively crawl semantic web links to construct multi-dimensional knowledge graphs. Extracted named entities, clustered contextual relationships, and mapped topical hierarchies using graph theory data structures, enabling researchers to visualize complex conceptual domains, uncover hidden nodal links, and explore interconnected data visually.

### Social Network Analysis (SNA) & Graph Centrality Metrics

Eigenvector Centrality Betweenness Degree Centrality  
Louvain Community Detection

Built network graph analytics platforms that reconstruct social communication structures and digital communities. Calculated eigenvector, degree, and betweenness centrality metrics to identify influential network nodes and information gatekeepers, applying Louvain community detection algorithms to segment topological clusters and analyze information propagation pathways across large datasets.



Modern React, Next.js & Strict TypeScript Architecture

React 18+   Next.js SSR/SSG   TypeScript   Tailwind CSS   Atomic Design Patterns

Architected responsive, production-ready web applications using Next.js, React, and TypeScript. Implemented server-side rendering (SSR), static site generation (SSG), and atomic UI component design with Tailwind CSS, ensuring strict end-to-end type safety, modular state management, and optimized asset delivery for high-performance responsive user experiences.

High-Throughput Asynchronous Python APIs

FastAPI   Pydantic   Uvicorn   Asyncio Event Loops   Connection Pooling   OpenAPI

Engineered high-concurrency backend services and RESTful API gateways using FastAPI and Pydantic. Leveraged asynchronous I/O loops ('asyncio'), database connection pooling, and strict schema validation models to deliver sub-millisecond response latencies under heavy traffic while automatically generating interactive OpenAPI (Swagger) documentation.

Enterprise CRM & Pipeline Telemetry Architecture

PostgreSQL   SQLAlchemy   Analytics Pipelines   Role-Based Dashboards   Relational Schemas

Developed full-stack CRM and enterprise telemetry systems featuring dynamic sales pipelines, lead lifecycle tracking, and interactive analytics. Designed normalized relational database schemas in PostgreSQL with SQLAlchemy ORM, implementing automated event triggers, activity audit histories, and role-based data filtering across enterprise organizational hierarchies.

Creative Web Development & Canvas Shader Animations

Three.js   WebGL Canvas   GSAP ScrollTrigger   99+ Lighthouse Optimization

Built award-caliber creative web interfaces integrating custom WebGL canvas shaders, Three.js 3D scenes, and GSAP ScrollTrigger animations. Engineered buttery-smooth micro-interactions, responsive fluid typography, and asset preloading strategies, consistently achieving 99+ Google Lighthouse performance, accessibility, and SEO audit scores.

Adaptive Video Streaming & Media Optimization

HLS / DASH Protocols   CDN Edge Caching   Video Transcoding   Lazy Loading

Built modern streaming frontends integrating adaptive bitrate video players, episode indexing, and dynamic metadata scrapers. Implemented CDN edge caching, video thumbnail generation, and client-side view state persistence, ensuring instant playback startup and seamless buffering across bandwidth-constrained mobile networks.





### Chrome Extension (Manifest v3) DOM Extraction

JavaScript Manifest v3 MutationObserver Content Scripts  
Background Workers

Developed high-performance Google Chrome extensions adhering to modern Manifest v3 standards. Leveraged DOM MutationObservers and asynchronous content scripts to extract dynamic real-time feed elements, social interaction metrics, and trending metadata, streaming structured JSON outputs locally without degrading browser tab rendering performance.

### Heuristic Human Emulation & Anti-Bot Automation

PyAutoGUI Selenium Bezier Curve Trajectories  
Micro-Pause Delay Jitter

Engineered sophisticated browser automation bots capable of simulating human interaction mechanics. Implemented cubic bezier mouse movement trajectories, non-linear velocity curves, randomized keystroke timings, and dynamic micro-pause jitter, reliably bypassing heuristic bot-detection engines and rate-limiting barriers during complex multi-page workflows.

### Multi-Egress Rotating Proxy & TLS Fingerprinting

SOCKS5/HTTP Proxies JA3/JA4 Spoofing  
Asynchronous Socket Pools IP Rotation

Built resilient network routing infrastructure managing dynamic pools of residential and datacenter proxies. Implemented automated IP rotation upon HTTP 429 throttling, health checking, and TLS client fingerprint spoofing (JA3/JA4 emulation), ensuring reliable, uninterrupted data acquisition across distributed web scraping and security assessment tasks.

### Automated Filesystem Hygiene & Daemon Scripting

Python OS/Shutil MIME Detection SHA-256 Hashing  
Windows Services Cron Daemons

Authored background filesystem automation services that monitor and organize local directories in real-time. Implemented MIME-type inspection, rule-based file taxonomy, and SHA-256 hash collision algorithms to automatically classify files, deduplicate storage, and maintain structured, tamper-evident archival logs across operating system environments.

### Digital Signal Processing (DSP) & MIDI Protocols

TypeScript Web Audio API Polyphonic Synthesizers  
Binary MIDI 1.0/2.0 Protocols

Developed comprehensive digital audio tools parsing binary MIDI 1.0/2.0 protocol specifications. Implemented raw byte decoding, multi-track timing clocks, velocity curve transformations, and real-time audio synthesis pipelines using the Web Audio API, delivering interactive piano-roll visualizations and high-fidelity polyphonic sound generation.

### Generative AI Document & Presentation Automation

python-pptx Structured Prompt Chains LLM Integration  
Layout Engines

Engineered automated presentation synthesis pipelines utilizing LLM prompt chains and 'python-pptx'. Programmatically generates structured slide hierarchies, contextual summaries, professional typography, and corporate color palettes directly from unstructured research documents, automating end-to-end slide deck generation without human manual formatting.